资料

launchd 服务排查小抄

一分钟排查

```
launchctl print gui/$(id -u)/com.example.app | head -30   # 状态与环境
launchctl kickstart -k gui/$(id -u)/com.example.app       # 重启
plutil -lint ~/Library/LaunchAgents/com.example.app.plist  # 语法
tail -20 ~/Library/Logs/app.log                           # 日志
```

最常见的三个坑

坑	现象	修法
PATH 缺 Homebrew	`command not found` / 功能静默失效	启动时补 PATH，或绝对路径 + `shutil.which`
工作目录不对	读不到相对路径文件	plist 配 `WorkingDirectory`
改完没重启	代码改了没生效	`launchctl kickstart -k`

plist 模板

```
<key>Label</key><string>com.example.app</string>
<key>ProgramArguments</key>
<array>
  <string>/opt/homebrew/bin/python3</string>
  <string>/path/to/app.py</string>
</array>
<key>WorkingDirectory</key><string>/path/to</string>
<key>EnvironmentVariables</key>
<dict><key>PATH</key><string>/opt/homebrew/bin:/usr/local/bin:/usr/bin:/bin</string>
        </dict>
<key>KeepAlive</key><true/>
<key>StandardOutPath</key><string>/Users/you/Library/Logs/app.log</string>
<key>StandardErrorPath</key><string>/Users/you/Library/Logs/app.log</string>
```

启动自检（避免静默失败）

```
for tool in ("pdftotext", "node", "ffmpeg"):
    log.info("依赖 %s: %s", tool, shutil.which(tool) or "缺失")
```

> 仅用于本人设备/自有服务。